# ENEL 585 – Lab 1

**DC-DC Conversion: Buck Converter (Open Loop) and Boost Converter (Closed-Loop Output Voltage Control)**

## Group Information

Group #: _____

Student name(s): _____

## Purpose

To operate one half-bridge leg of the inverter box as two different DC-DC converters. In Part A you will run a buck converter in open loop and connect its measured behaviour (conversion ratio, inductor current ripple, conduction modes) to the analysis from class. In Part B you will re-wire the same leg as a boost converter and operate it under closed-loop output voltage control, observing what feedback does that open-loop operation cannot. You will also use the PLECS-to-LaunchPad workflow (build, flash, External Mode) that you will rely on for the final project.

## Marking

**Total Marks for Pre-Lab Quiz = __ /10 (Weight = 2%)**

**Total Marks for Lab Report = __ /10 (Weight = 2%)**

**Total Marks for Lab Work = __ /10 (Weight = 2%)**

The pre-lab itself is not collected or graded. Instead, a **10-minute individual quiz** is written at the start of the lab session, based directly on the pre-lab questions below. The lab report is marked against the rubric in the Lab Report section, not experiment by experiment.

*Revised Sep. 13 after the first lab session: content that changed from the Sep. 10 version is highlighted like this.*

## Associated Documents

The orientation lab covered PLECS, the LaunchPad and External Mode — this handout assumes you can build, flash and connect to the LaunchPad without instructions. What is **new** is the inverter box: the orientation lab used the Typhoon HIL, not this hardware, so the Part A setup below walks the box wiring in order and points to the guide sections that carry the detail. Read the guides before the session and allow about 30 minutes for the first setup and its sign-off.

- *PLECS Setup* — installing PLECS and the TI C2000 TSP, building and flashing the LaunchPad, using External Mode (live scopes and parameter tuning).
- *Inverter Box Setup Guide* — wiring the inverter box, interface board and LaunchPad: auxiliary +15 V power, thermal trip, gate-drive BNCs, probe inputs and their scaling.
- *Bench Instruments Guide* — front-panel procedures for the E3631A, BK 1666 and ITECH IT-M3632 supplies and the oscilloscope and probes.
- PLECS model files (on D2L): buck_open_loop.plecs (Part A) and boost_closed_loop.plecs (Part B).

## Introduction

Fig. 1 shows the buck converter you will build in Part A from one leg of the inverter box. Both devices of the leg are IGBTs with antiparallel diodes, and both are always physically present. In Fig. 1a only TOP1 is gated: the BOT1 gate BNC is left disconnected, so BOT1 never conducts as a transistor and its antiparallel diode carries the freewheeling current — the leg is a conventional buck with a diode. In Fig. 1b both devices are gated complementarily (synchronous operation). You will operate both configurations.

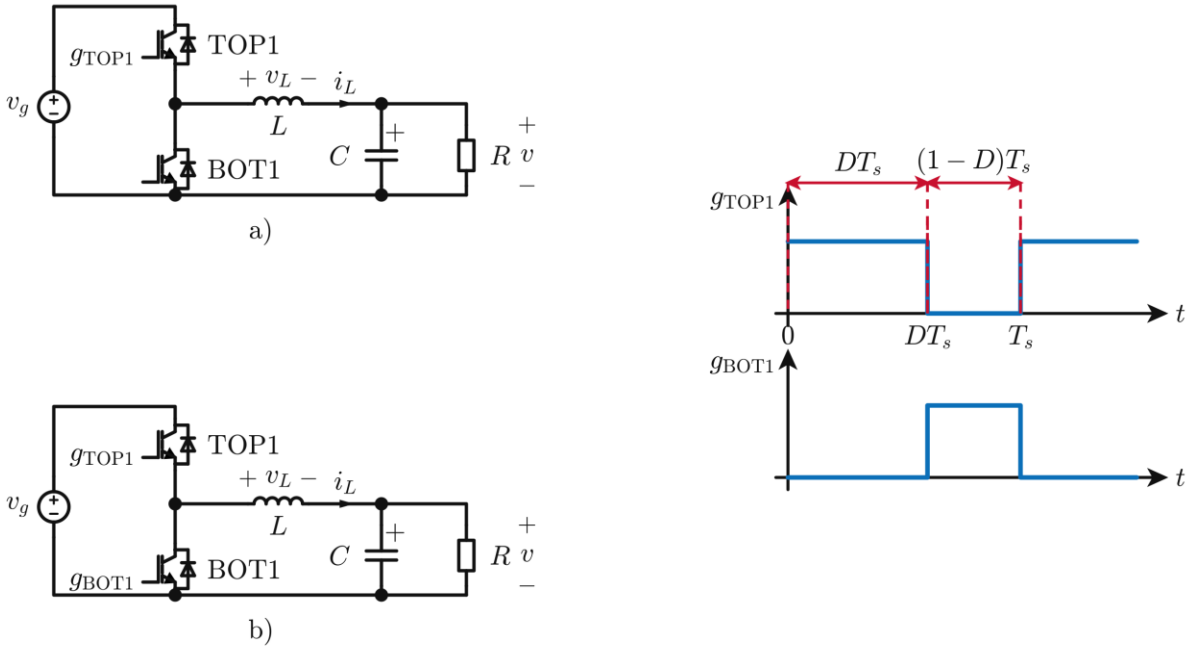



**Fig. 1** – Buck converter formed from one inverter leg. a) BOT1 ungated: its antiparallel diode freewheels. b) BOT1 gated complementarily to TOP1. The gate traces apply to b); dead time is omitted.

Fig. 2 shows the Part B circuit. It is the **same half-bridge leg** with the direction of power flow reversed: the supply now feeds the inductor into the leg midpoint, and the output is taken across the DC link. BOT1 is the active (boost) switch and its on-time fraction is the duty ratio *D*. TOP1 is not gated at all in Part B: its antiparallel diode is the boost rectifier. No external capacitor is needed — the inverter box has $C_{dc}$ = 1.1 mF permanently connected across its DC link, and that capacitor is the boost output filter.

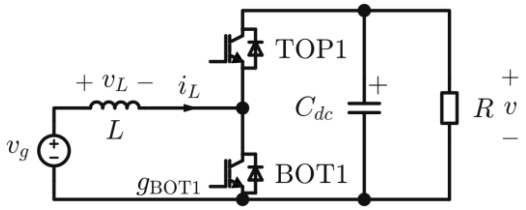



**Fig. 2** – Boost converter formed from the same leg: supply and inductor at the midpoint side, load across the DC link. TOP1 is ungated and works as the boost diode.

In Part B the duty ratio is not set by you: a **pre-designed PI controller** measures the output voltage and adjusts the duty ratio to track a reference you set from PLECS. You are not expected to design this controller — converter dynamics and control design come later in the course, and you will design loops like this one for the final project. Three limits are built in and are not to be modified: the reference is restricted to 20–45 V in the PLECS model, the duty ratio is clamped to $D \leq 0.7$, and the firmware trips the gating if the sensed output voltage exceeds 55 V.

## Pre-Lab

Work these on paper before the session; the pre-lab quiz at the start of the lab is drawn from them. Assume ideal components throughout. **Alongside each sketch, state the calculated average output voltage, average inductor current, duty ratio where applicable, and peak-to-peak inductor current ripple** — these are the predictions your report compares against.

### Part A — Buck (Fig. 1a)

Parameters: $v_g = 40.0$ V, $L = 5$ mH, $C = 120$ µF, $f_s = 1$ kHz. The load is the ITECH electronic load in constant-resistance mode; R is given in each question.

1. For $D = 0.8$ and $R = 8\ \Omega$: sketch $v_L(t)$ and $i_L(t)$ over one switching period, labelling the voltage levels, the current extremes, and the time instants at which the switching state changes.
2. Repeat question 1 for $D = 0.2$ and $R = 25\ \Omega$. Verify whether the continuous-conduction assumption underlying your analysis is satisfied at this operating point, using the ripple and average current you calculated. If it is not, identify the conduction mode that results, sketch the corresponding $v_L(t)$ and $i_L(t)$, and state which device conducts in each interval.
3. At $D = 0.8$, derive the load current at which the converter reaches the boundary between CCM and DCM, and the equivalent load resistance. State the condition on the inductor current that defines the boundary.

### Part B — Boost (Fig. 2)

Parameters: $v_g = 20.0$ V, $R = 12.5\ \Omega$, $L = 5$ mH, $C_{dc} = 1.1$ mF, $f_s = 5$ kHz. TOP1 is ungated (Fig. 2).

1. For an output voltage of 40 V: sketch $v_L(t)$ and $i_L(t)$ over one switching period, labelling the voltage levels, the current extremes, and the time instants at which the switching state changes, and naming the device that conducts in each interval.
2. The ideal conversion ratio of the boost converter is $M(D) = 1/(1-D)$. (a) State the limit this expression predicts as $D \to 1$, and identify the non-ideal elements of a real converter that prevent it from being reached. (b) For a 20 V input, determine the output voltage and load current that the firmware duty limit $D = 0.7$ would ideally produce into 12.5 Ω. (c) Explain the role of the duty-ratio limit as a protection in a closed-loop boost converter, with reference to the case in which the controller demands an output voltage the converter cannot deliver.
3. The load resistor must remain connected during boost-converter operation. Analyze the effect of sudden load disconnection during steady-state voltage regulation. Using capacitor current balance, explain the initial output-voltage response before the controller significantly changes the duty ratio. Describe the controller's subsequent corrective action and explain why voltage feedback alone does not guarantee overvoltage protection.

## Safety Notes

- The DC link stores energy after shutdown. The internal 1.1 mF capacitor discharges through balancing resistors with a time constant of roughly 46 s when no other discharge path is present. After switching off, measure the DC link and the external capacitor block and confirm both are below 10 V before touching or re-wiring the circuit.
- Make every wiring change with all supplies off and switching disabled. Never re-wire an energized circuit.
- The external capacitor block used in Part A is **polarized** — reverse polarity will damage it. Its red terminal goes toward the inductor.
- Set the specified supply and load limits before connecting anything, and obtain TA sign-off before applying power. Switching is controlled from enableswitch in External Mode, which selects the disable (0) or enable (1) constant — do not edit the constants. Select disable before any supply is turned on, and enable only once the input is at its operating voltage. In Part A the ITECH load is turned on last, at 100 Ω, and stepped down to the required resistance. At shutdown: select disable, reduce the input voltage to minimum and switch the supply output off (the BK 1666 has only its power switch), then turn the load off. Verify discharge and turn off the auxiliary supply before changing any wiring.

## Equipment

| Component | Purpose | Qty |
|---|---|---|
| Oscilloscope | Display probed signals | 1 |
| Keysight 1146B clamp-on current probe | Measure iL | 1 |
| Keysight N2791A differential voltage probe | Measure v | 1 |
| BNC splitter | Share the voltage-probe signal between the interface board and the oscilloscope | 1 |
| Inverter box (SEMITEACH) | Half-bridge leg (TOP1/BOT1) and internal 1.1 mF DC-link capacitor | 1 |
| Interface board + LAUNCHXL-F28P55X | Gate drive, sensing and control (see Inverter Box Setup Guide) | 1 |
| Laptop with PLECS + TI C2000 TSP | Build, flash and monitor the controller (see PLECS Setup) | 1 |
| 5 mH inductor | L in Figs. 1 and 2 | 1 |
| 120 µF capacitor block | C in Fig. 1 (Part A only) | 1 |
| Resistor load bank | R in Fig. 2 (Part B): 20 × 250 Ω in parallel, one switch each — 1 switch = 250 Ω, 5 = 50 Ω, 10 = 25 Ω, all 20 = 12.5 Ω | 1 |
| Keysight/Agilent E3631A supply | +15 V auxiliary power for gate drivers and interface board | 1 |
| BK Precision 1666 supply | Part A input source (40 V) | 1 |
| ITECH IT-M3632 | Part A electronic load (constant-resistance mode); Part B input source (20 V) | 1 |
| Banana and BNC cables | Wiring per the setup guides | — |

# Instrument Settings for Lab 1

Front-panel procedures for every instrument are in the *Bench Instruments Guide* — read it before the session. Set every limit before the instrument is wired into the circuit, with its output off. Lab 1 uses these settings; the TA checks them at sign-off:

- E3631A: +25V output at 15.00 V with a 1.000 A limit (Guide §1).
- **BK 1666 (Part A input):** both Current knobs fully clockwise (5 A limit); both Voltage knobs fully counter-clockwise until startup. It has no output on/off key — "BK off" means its power switch (Guide §2).
- **ITECH IT-M3632:** Part A — load mode, constant resistance, starting at 100 Ω. Part B — source mode, 20 V, 10 A current limit, OVP 25 V (Guide §3).
- **Oscilloscope:** set it up by hand — do not use [Auto Scale] — and read averages and peak-to-peak values with [Meas] (Guide §4). Voltage probe 10:1; current probe 100 mV/A. Starting settings are in Table 1.
- Experiment 6 capture: trigger on v (rising, level ≈27 V, just above the 25 V starting value) in Normal mode, move the trigger point about two divisions from the left edge, press [Single], then step the reference. The response takes a few hundred milliseconds to settle.

**Table 1: Suggested starting oscilloscope settings — adjust as needed**

| Experiment | Time/div | v channel | iL channel | Trigger |
|---|---|---|---|---|
| 1: buck, D = 0.8, 8 Ω | 200 µs | 5 V/div, 0 V at bottom | 1 A/div, 0 A at bottom | iL, rising, Auto |
| 2: buck boundary, 8 → 100 Ω | 200 µs | 5 V/div, 0 V at bottom | 1 A/div, down to 200 mA/div near boundary | iL, rising, Auto |
| 3: buck DCM, D = 0.2, 25 Ω | 200 µs | 2 V/div, 0 V at bottom | 200 mA/div, 0 A at bottom | iL, rising, Auto |
| 4: synchronous, D = 0.2, 25 Ω | 200 µs | 2 V/div, 0 V at bottom | 500 mA/div, 0 A at centre | iL, rising, Auto |
| 5 and 7: boost, 40 V | 50 µs | 10 V/div, 0 V at bottom | 1 A/div DC for the average; AC, 100 mA/div for ripple and D | iL, rising, Auto |
| 6: reference step 25 → 45 V | 50 ms, trigger point 2 div from left | 10 V/div, 0 V at bottom | 2 A/div, 0 A at bottom | v, rising, ≈27 V, Normal + [Single] |

# Part A Setup — Buck Converter

You have not used the inverter box before this session, so the hardware steps are in order and point to the guide sections that carry the detail. Allow about 30 minutes; the TA checks the wiring before power is applied.

### Hardware (Fig. 1a)

1. Control side, per the *Inverter Box Setup Guide* (Setup Procedure, then the Interface Board sections): seat the LaunchPad on the interface board, connect the +15 V auxiliary supply, wire the thermal-trip loop and the ERROR1 BNC, then connect the **TOP1 gate BNC only**. Leave the BOT1 gate BNC disconnected until Experiment 4.

2. Power circuit of Fig. 1a with banana cables: BK 1666 across the inverter DC-link terminals (mind polarity — the internal 1.1 mF capacitor sits here), then leg midpoint → 5 mH inductor → capacitor block ‖ ITECH load input. The capacitor block's red terminal goes toward the inductor. Leave the BK 1666 switched off, with both Current knobs fully clockwise (5 A limit) and both Voltage knobs fully counter-clockwise (see Bench Instruments Guide §2).
3. ITECH as load: select load mode and constant resistance (CR), set 100 Ω, and leave the input off. After startup, decrease resistance through 50, 25, 12.5 and 8 Ω, allowing the readings to settle at each step. Decreasing resistance increases loading. Confirm load mode on the display; the same instrument becomes the source in Part B. Key sequence: Bench Instruments Guide §3.
4. Voltage probe (set 10:1) across the load; via the BNC splitter, feed it to both VProbe1 on the interface board and the oscilloscope. Current probe (set 100 mV/A, zeroed) on the inductor wire, to the oscilloscope.
5. BK 1666: switched off, both Voltage knobs fully counter-clockwise, both Current knobs fully clockwise (5 A limit). E3631A: +15 V, 1 A limit, output on; check that the interface-board supply LEDs D5, D9 and D4 are lit.

**LaunchPad power:** the LaunchPads on the inverter-box benches are in the experimental set-up (JP1 open; see PLECS Setup, section 3), so they take their power from the interface board, not from the USB cable. **The E3631A +15 V supply must be on — D5, D9 and D4 lit — before you build and flash or connect External Mode;** with it off, flashing and External Mode fail even though the USB cable is plugged in. Turning the +15 V supply off resets the LaunchPad and the program restarts with its built-in values: after turning it back on, reconnect External Mode, check that enableswitch selects disable, and re-enter the duty ratio before continuing. Do not change the LaunchPad's jumpers or switches.

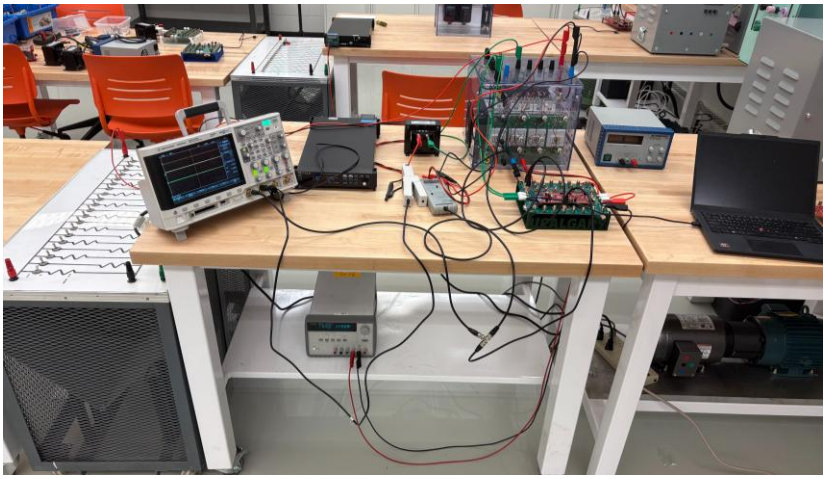


**Fig. 3** – Part A buck converter setup (Fig. 1a).

### Software (buck_open_loop.plecs)

1. Open buck_open_loop.plecs and build/flash it per PLECS Setup §4 (target TI28P55x, Build and program, Run from Flash). The +15 V supply must be on first (see LaunchPad power, above).
2. Connect External Mode per PLECS Setup §5. The duty ratio is a tunable parameter; the model scope shows the sensed output voltage.
3. **Note:** the output voltage shown in PLECS will differ from the oscilloscope reading by a few volts. This is a known offset in the interface-board sensing hardware, not a wiring fault; the oscilloscope is the reference measurement, and the controller in Part B regulates the sensed value. Record both readings whenever you report an output voltage.

4. Leave enableswitch selecting disable until the TA has verified the circuit. Startup order: +15 V auxiliary power on and status LEDs checked → BK switched on and raised to 40 V → select enable → ITECH load on at 100 Ω → decrease resistance to the value the experiment specifies.

**STOP:** TA to verify the circuit before the 40 V supply is turned on.

## Part A — Buck Converter, Open Loop

All Part A experiments: $v_g$ = 40 V, $f_s$ = 1 kHz, ITECH load in CR mode at the R given in each table. For each experiment, capture the v and $i_L$ waveforms with the oscilloscope (label which trace is which) and answer the questions in your report. Set up the oscilloscope by hand (Table 1; no [Auto Scale]) and read averages and peak-to-peak values with its [Meas] functions.

### Experiment 1

**Table 2: Experiment 1 parameters**

| Bottom switch | Control | R [Ω] | fs [Hz] | D |
|---|---|---|---|---|
| Diode (BOT1 BNC disconnected) | Open loop | 8 | 1000 | 0.8 |

**Questions:**

- Measure the average and peak-to-peak inductor current, and the average output voltage. Compare with your pre-lab values and account for any deviation.
- Is the converter in CCM or DCM? Justify from the waveform.
- Compute the power transfer efficiency from the BK front-panel readings and the ITECH's voltage and current readings; this excludes the separately supplied auxiliary power. Identify the loss mechanisms.

### Experiment 2

Keep D = 0.8 with BOT1 ungated, starting from R = 8 Ω as in Experiment 1. Raise the ITECH resistance in steps (8 → 12.5 → 25 → 50 → 100 Ω), watching the inductor current, until a zero-current interval appears; refine R to the boundary.

**Questions:**

- Report the load current and equivalent resistance at the boundary and compare with your pre-lab prediction. What does the boundary current tell you directly about the ripple?
- Capture the waveform that shows the converter is at the boundary.

### Experiment 3

**Table 3: Experiment 3 parameters**

| Bottom switch | Control | R [Ω] | fs [Hz] | D |
|---|---|---|---|---|
| Diode | Open loop | 25 | 1000 | 0.2 |

Set the ITECH resistance to 100 Ω and D = 0.2 in External Mode, then decrease the resistance through 50 Ω to 25 Ω. BOT1 stays ungated.

**Questions:**

- Measure the average output voltage. It will not equal $Dv_g$ — explain why.
- Annotate the intervals of the inductor current waveform: which device conducts in each, and why does the current stay at zero for part of the period?

## Experiment 4

**Table 4: Experiment 4 parameters**

| Bottom switch | Control | R [Ω] | fs [Hz] | D |
|---|---|---|---|---|
| IGBT (BOT1 BNC connected) | Open loop | 25 | 1000 | 0.2 |

Select disable using enableswitch, turn the BK Voltage knobs fully counter-clockwise and switch it off, then turn the ITECH load off. **Now connect the second gate BNC:** a BNC cable from the Bot 1 gate-drive output on the interface board to the Bot 1 gate input on the inverter box (labels match on both ends). Without it BOT1 stays ungated and the waveforms will look the same as in Experiment 3. Keep D = 0.2, set the ITECH resistance to 100 Ω, then follow the startup sequence and decrease resistance through 50 Ω to 25 Ω.

**Questions:**

- Compare the waveforms and average output voltage with Experiment 3. What changed, and why does gating BOT1 eliminate the behaviour you saw there?
- Identify the intervals of negative inductor current. For each gating interval, identify the conducting device for positive and negative current (TOP1 IGBT, TOP1 diode, BOT1 IGBT or BOT1 diode). An IGBT conducts in one direction only, regardless of its gate command.

## Re-configuring for Part B

1. Select disable using enableswitch, turn the BK Voltage knobs fully counter-clockwise and switch it off, then turn the ITECH load off.
2. Measure the DC link and external capacitor block; wait until both are below 10 V before re-wiring. Turn off the +15 V auxiliary supply before changing power or gate-command connections.
3. Re-wire to **Fig. 2**: ITECH → 5 mH inductor → leg midpoint; resistor load bank across the DC-link terminals with **one switch closed** (250 Ω). The bank is 20 × 250 Ω in parallel: closing more switches lowers the resistance. The BK 1666 and the capacitor block are no longer used. **Disconnect the TOP1 gate BNC** — only BOT1 is gated in Part B.
4. Switch the ITECH to source mode and configure it before connecting its output: 20 V, current limit 10 A, OVP 25 V, output off. The ITECH OVP monitors the boost input and does not protect the DC-link output. Keep the load connected and retain all prescribed firmware protections. Key sequence: Bench Instruments Guide §3.
5. With power-stage outputs off and enableswitch selecting disable, restore +15 V auxiliary power, check the status LEDs, and flash boost_closed_loop.plecs. Reconnect External Mode. After TA sign-off, turn on the ITECH source, compare the sensed output voltage with the oscilloscope reading, set the reference to 25 V, select enable, then raise the reference to 40 V. Only then close the remaining switches to reach 12.5 Ω (all 20 closed). (The LaunchPad restarts when +15 V is restored — see LaunchPad power in Part A Setup.)



**Fig. 4** – Part B boost converter setup (Fig. 2).

**STOP:** TA to verify the re-wired circuit before the ITECH output is turned on.

## Part B — Boost Converter, Closed-Loop Output Voltage Control

All Part B experiments: $v_g$ = 20 V, $f_s$ = 5 kHz, closed-loop control active. The output voltage reference is set from External Mode (20–45 V). Keep the load bank connected at all times while switching — never open every switch; change the resistance by closing or opening switches one at a time with the others still closed. Use the oscilloscope settings in Table 1 and its [Meas] functions.

## Experiment 5

**Table 5: Experiment 5 parameters**

| Control | R [Ω] | fs [Hz] | Vo reference [V] |
|---|---|---|---|
| Closed loop | 12.5 | 5000 | 40 |

**Questions:**

- Capture the steady-state waveforms. Measure the average output voltage, the average inductor current and the peak-to-peak ripple; compare with your pre-lab values.
- Determine the BOT1 duty ratio from the inductor current waveform: the current rises while BOT1 conducts and falls while TOP1's diode conducts, so D is the rising interval divided by the switching period. Check it against the slopes, $v_g/L$ and $(v_g - v)/L$. Compare with the ideal prediction and explain the difference using the circuit's non-ideal behaviour and your measurements.
- Estimate the efficiency: input power from the ITECH voltage and current readings, output power as $V_{rms}^2/R$ (or state the small-ripple approximation if you use the average voltage). As in Part A, this estimate excludes the separately supplied auxiliary power. Relate the efficiency to the duty-ratio difference above.

## Experiment 6

With the converter regulating at 25 V, capture the output voltage and inductor current on the oscilloscope during a reference step from 25 V to 45 V. Return the reference to 40 V afterwards. At 45 V the input current is about 9 A; note whether the 10 A source limit engages.

**Questions:**

- Report the rise time, overshoot and settling time of the output voltage. Measure rise time from 10% to 90% of the voltage change; report overshoot relative to the final reference and settling time to a ±2% band around it.
- From the inductor current, describe what the duty ratio does during the transient. Why must it temporarily differ from its final steady-state value? How close does it get to the 0.7 clamp at 45 V, and what would happen to the response if the step were larger?

## Experiment 7

**Table 6: Experiment 7 parameters**

| Control | R [Ω] | fs [Hz] | Vo reference [V] |
|---|---|---|---|
| Closed loop | 12.5, 25, 50 | 5000 | 40 |

With the converter regulating at 40 V, record the steady-state output voltage and inductor current waveform at R = 12.5 Ω (all 20 switches), 25 Ω (10 switches) and 50 Ω (5 switches). Open switches one at a time to go from one setting to the next; no transient capture is required.

**Questions:**

- Tabulate the average output voltage, the average inductor current, the peak-to-peak ripple and the duty ratio at the three loads. The average current changes with load while the ripple and the duty ratio do not. Explain why, from the inductor volt-second balance and the boost conversion ratio.
- Would the output voltage stay at 40 V across these three loads if the loop were disabled and the duty ratio frozen? Answer for the ideal converter and then for the real one, naming the effect that separates them. (Answer from theory — do not disable the loop.)

## Shutdown

1. Reduce the reference to 20 V and select disable using enableswitch. Then ramp the ITECH voltage down to zero before turning its output off — the ramp discharges the DC link through the source. Leave the resistor load connected.
2. Measure the DC link and confirm it has discharged below 10 V.
3. Turn off the E3631A, disconnect your wiring, and return the lab equipment as instructed by the TA.

## Lab Report

Submit one report per group on D2L by 10:00 AM one week after your lab session. The report is marked out of 10 using the rubric below, rather than experiment by experiment. Include the required captures and measurements, answer the questions directly, and use your results to support your explanations. Do not narrate the procedure.

Table 7: Lab report rubric (10 marks)

| Criterion | Marks | Full-marks standard |
|---|---|---|
| Experimental evidence | 3 | Required captures and measurements are present and usable. Channels, scales, zero levels, probe factors, operating conditions and units are clear where applicable. Relevant measurement limitations are identified. |
| Analysis and comparison | 3 | Calculations are correct. Measurements are compared quantitatively with applicable theoretical predictions, with assumptions stated. Relevant trends across operating conditions are identified. |
| Physical interpretation | 4 | Answers explain the observed behaviour using switching states, current paths and circuit principles. Significant discrepancies are discussed using the available evidence, with uncertainty acknowledged. |

Table 8: Quality scale, applied separately to each criterion

| Level | Share of marks | Standard |
|---|---|---|
| Excellent | 90–100% | Complete and technically correct; no substantive weaknesses. |
| Good | 75–89% | Substantially complete and correct; minor weaknesses do not undermine the results or conclusions. |
| Adequate | 60–74% | Demonstrates the essential requirements, but omissions or errors limit the analysis, interpretation or usefulness of the evidence. |
| Marginal | 50–59% | Provides some relevant evidence or understanding, but substantial omissions or errors undermine the results or conclusions. |
| Inadequate | below 50% | Provides little usable evidence of meeting the criterion, or contains fundamental errors. No relevant work earns zero. |